

Compress and Forget: bitsandbytes Quantization Amplifies Proactive Interference in LLMs

Shayan Shahrabi-Farahani

Shahid Beheshti University, Tehran, Iran
sh.shahrabifarahani@mail.sbu.ac.ir

Dara Rahmati

Shahid Beheshti University, Tehran, Iran
d.rahmati@sbu.ac.ir

Abstract

Proactive Interference (PI) is a documented LLM failure mode in which retrieval of a repeatedly overwritten value degrades as prior overwrites accumulate, mirroring a classical phenomenon in human working memory. Post-training quantization (PTQ) is now the default deployment path for open-weight models, yet its effect on this failure mode has not been tested. We evaluate three precision levels (FP16, INT8, INT4/NF4, via bitsandbytes) across three architecturally distinct instruction-tuned models (Qwen2.5-7B-Instruct, Mistral-7B-Instruct-v0.3, Phi-3.5-mini-instruct), holding the retrieval task fixed. It turns out that INT4 quantization significantly reduces accuracy under high interference in every model (e.g., 81.0% $\rightarrow$ 68.3% for Qwen), confirmed by paired McNemar’s tests ($p \leq 2.6 \times 10^{-6}$) and a mixed-effects regression spanning all interference levels; INT8, often assumed safe, also carries a smaller but real penalty in two of three models. The effect is specific to semantically similar (word-type) distractors and reverses sign under a numeric control condition, and is mechanistically linked to a rise in same-key intrusion errors under INT4 (21.5% $\rightarrow$ 24.6% of trials, $p = 4.8 \times 10^{-7}$). A follow-up ablation shows the effect originates in the quantized transformer backbone rather than the output projection layer. These results suggest that bitsandbytes 4-bit quantization can impose an additional cost on applications relying on long, updatable, semantically dense contexts, even when aggregate benchmark accuracy appears largely unaffected.

We release our code and tokenizer-verified vocabulary construction method at <https://github.com/ShayanShahrabi/compress-and-forget>

1 Introduction

Post-training quantization (PTQ) is the default path by which most practitioners deploy open-weight LLMs. 8-bit and 4-bit weight quantization via libraries such as bitsandbytes [1, 2], GPTQ [3], and AWQ [4] routinely report negligible loss on aggregate benchmarks (perplexity, MMLU, standard QA). This has produced a general assumption that 4-bit quantization is *safe enough* for most purposes.

Separately, a growing line of work shows that LLMs are systematically vulnerable to *proactive interference* (PI); when a value is updated multiple times within a context (for example, a meeting time that is first set to 2pm, then moved to 3pm, then moved again to 4pm), retrieval of the final value degrades log-linearly as the number of prior updates grows, even though the correct value is the one nearest the query [5]. This failure mode is distinct from long-context retrieval failure in the *needle-in-a-haystack* sense [6]: the interfering values are short, few in number relative to full context windows, and the task provides no ambiguity about which value is being asked for. A recent large-scale study across 39 models further shows that PI is a stronger and more universal failure mode in transformers than the symmetric case (retroactive interference), and that it reflects a distinct, capacity-independent mechanism – a bias toward protecting early information at the expense of recent updates [7].

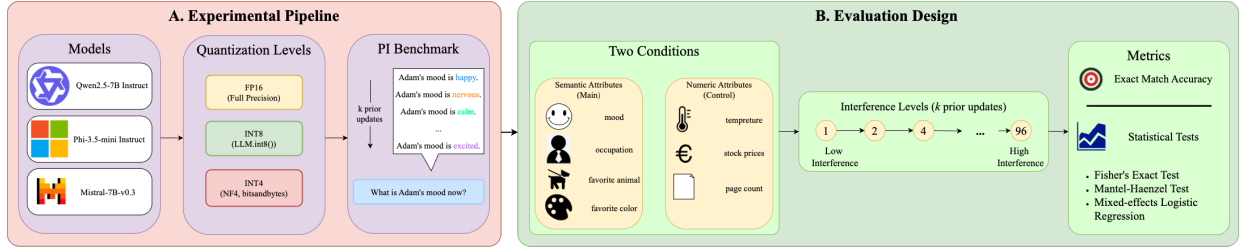


Figure 1: Overview of the experimental pipeline and evaluation design. (A) Each of three instruction-tuned models is loaded at three precisions (FP16, INT8, INT4/NF4 via bitsandbytes) and evaluated on the PI-LLM retrieval benchmark, in which a subject’s attribute is overwritten k times and the model is queried for only the most recent value. (B) Two attribute conditions are tested in parallel – semantic (word-type) attributes as the main condition and numeric attributes as a control – across interference levels from $k = 1$ to $k = 96$, with exact-match accuracy and paired statistical tests (Fisher’s exact, Mantel–Haenszel, mixed-effects logistic regression) computed at each level.

These two lines of work have not been connected. It is not known whether the aggregate-benchmark robustness of quantized models extends to interference-heavy retrieval, or whether compressing model weights degrades this specific, already-fragile capability disproportionately. This matters practically, where applications with long-lived, frequently updated states (multi-turn assistants, document revision tracking, dialogue systems that track user preferences) are exactly the setting where PI is most likely to occur, and are also the setting most likely to be served by quantized models for cost reasons.

We test this directly. Holding the PI-LLM retrieval task fixed, we vary only quantization precision (FP16 / INT8 / INT4) across three open, architecturally distinct 4–7B instruction-tuned models. We find a consistent, statistically significant effect: 4-bit quantization specifically degrades retrieval under semantic interference (e.g. mood, occupation, favorite color/animal attributes), while an arbitrary numeric-attribute control condition shows no such effect. This specificity is the central finding of the research. It shows that quantization is not simply *noisier* in a way that hurts all retrieval equally – it selectively erodes the mechanism that resists semantically confusable distractors. Figure 1 present an overview of what the procedure looks like.

Our main contributions are summarized as follows:

- We provide an initial systematic evaluation of how post-training quantization affects proactive interference in LLMs, examining three precision levels across three architecturally distinct open-source models.
- We show the effect is specific to semantic (word-type) interference and does not generalize to arbitrary (numeric) interference, ruling out a generic noise-injection account.
- We identify a *cliff pattern* rather than a graceful degradation: INT8 is statistically indistinguishable from FP16 in two of three models, while INT4 shows a significant, consistent drop – suggesting the effect is not simply proportional to quantization aggressiveness.
- We release our task-generation code, tokenizer-verified vocabulary construction method, and full trial-level results.

2 Related Work

Proactive interference in LLMs. Wang and Sun [5] introduce PI-LLM, adapting the classical proactive

interference paradigm from cognitive psychology: a key is repeatedly rebound to new values within a context, and the model is queried for only the final value. Despite the target being unambiguous and recency-adjacent, retrieval accuracy declines log-linearly as the number of prior rebindings grows; the authors show this holds across model families and is not resolved by prompting the model to ignore earlier values, nor reliably by chain-of-thought reasoning. Chattaraj and Raj [7] extend this by directly contrasting proactive and retroactive interference across 39 models, finding PI dominates RI universally (Cohen’s $d = 1.73$) – the opposite of the typical human pattern – and that PI resistance is uncorrelated with model scale, unlike RI resistance. This suggests PI reflects a structural property of transformer attention (a primacy bias) rather than a capacity limitation that scales away with larger models. Xie [8] propose a sleep-inspired memory consolidation mechanism to mitigate PI, framing it as an learned, selective-forgetting problem analogous to biological memory consolidation. None of this line of work examines quantization as a variable.

Post-training quantization. LLM.int8() [1] and QLoRA [2] established that 8-bit and 4-bit (NF4) post-training quantization preserve aggregate task performance with minimal loss. GPTQ [3] and AWQ [4] improve on accuracy-per-bit through more sophisticated calibration, but the underlying evaluation methodology in this literature is near-universally aggregate benchmark accuracy (perplexity, MMLU-style QA), not fine-grained behavioral failure modes. A recent large-scale benchmarking effort, PTQ-Bench [9], systematically taxonomizes weight-only PTQ strategies (compensation-based, optimization-based, rotation-based, and salience-based) and evaluates their cross-bitwidth, cross-architecture, and cross-modality robustness across model sizes from 7B to 70B; notably, its recommendations are again derived entirely from aggregate task metrics rather than from behavioral probes of a specific cognitive-style failure mode such as PI, and it does not evaluate bitsandbytes-style quantization specifically. This gap – broad, systematic PTQ benchmarking that nonetheless stays at the aggregate-metric level – is precisely what motivates our narrower but behaviorally-targeted evaluation. Separately, Zhang et al. [10] study quantization’s effect on *catastrophic forgetting during continual fine-tuning*, finding – counterintuitively – that 8-bit quantization noise acts as an implicit regularizer that *improves* retention relative to FP16, while 4-bit quantization is more sensitive to replay buffer size. This is a training-time phenomenon (weights are updated sequentially across tasks); our work is purely inference-time (weights are frozen, only in-context retrieval is tested), making the two effects mechanistically distinct despite superficial similarity in framing. Zhang et al. [11] show a related but distinct interaction: quantization can causally resurface knowledge that fine-tuning-based unlearning had suppressed, again pointing to quantization interacting with a model’s information-handling behavior in ways aggregate benchmarks do not surface. Outside the language-model setting, Fu et al. [12] present an empirical study of quantization in visual world models (DINO-WM), showing that quantization-induced failures in planning tasks are task-specific and asymmetric across model components (encoder vs. predictor) rather than a simple function of bitwidth. While this work evaluates a different model class and modality entirely, it lends independent, cross-domain support to a theme our results also suggest: aggregate bitwidth-vs-accuracy summaries can obscure component- or task-specific quantization failure modes that only appear under targeted behavioral evaluation.

Long-context retrieval. Liu et al. [6] show LLMs exhibit position-dependent retrieval accuracy over long contexts (lost in the middle), a distinct phenomenon from PI: their setting involves a single unambiguous target among irrelevant distractors at varying positions, not a repeatedly overwritten value. Our prompts are short relative to typical long-context evaluations (at most a few hundred tokens even at our highest interference level), so our effect is not attributable to position-in-context retrieval degradation of the kind Liu et al. document. More directly relevant, Mekala et al. [13] systematically evaluate several post-training quantization methods on long-context tasks and report that low-bit weight-only quantization, including bitsandbytes’ NF4 scheme, is disproportionately damaging in retrieval-heavy long-context settings compared to other 4-

bit quantizers tested. Although their tasks involve much longer contexts and a different retrieval paradigm (no repeated key-rebinding) than ours, their finding that NF4 specifically – rather than 4-bit quantization in general – is among the worst-performing methods for retrieval-heavy tasks converges with our own restriction to bitsandbytes and strengthens the case that the effect we report is not an idiosyncrasy of our particular task design.

3 Method

This section gives a brief overview of the task, models, and analysis approach; full implementation detail (exact prompts, scoring rules, vocabulary construction, reproducibility configuration, trial budgets, and statistical formulas) is deferred to the Appendix section.

3.1 Task design

We adapt the PI-LLM key-rebinding paradigm from [5] where a named subject’s attribute is updated k times in sequence, defining the interference level, and the model must retrieve only the final value, disregarding the $k - 1$ overwritten prior values. We test eight attributes split into two conditions across interference levels $k \in \{1, 2, 4, 8, 16, 32, 64, 96\}$. Four **word-type (semantic)** attributes (mood, favorite color, favorite animal, occupation), where distractors are drawn from a shared semantic category, and three **numeric (control)** attributes (temperature, stock price, page count), where distractors are arbitrary integers with minimal semantic confusability. Prompts are passed through each model’s native chat template as a single user message with no system prompt, and models are instructed to answer with a single word or number. Exact prompt templates, scoring criteria, and our tokenizer-verified vocabulary construction procedure are given in Appendix A.

3.2 Models and quantization

We evaluate three open-weight, instruction-tuned models spanning distinct training organizations and architectures: Qwen2.5-7B-Instruct [14], Mistral-7B-Instruct-v0.3 [15], and Phi-3.5-mini-instruct [16]. Each is loaded at three precisions via the bitsandbytes library [1, 2]: FP16 (native), INT8 (default LLM.int8() decomposition), and INT4 (NF4 with double quantization). No layers were explicitly excluded from quantization for any arm, including the language modeling head (`lm_head`). A follow-up ablation (Section 4.7) found, however, that under this library version, the default INT4 loading path leaves `lm_head` at full precision regardless – confirmed by inspecting the loaded module directly. All INT4 results reported in this paper are therefore *backbone-only* quantization; Section 4.7 shows this does not affect our conclusions. Generation is greedy throughout (deterministic given a prompt); the only source of trial-to-trial randomness is the sampled subject name and distractor values, controlled by an explicit random seed. Exact loading configuration and package versions, provided for reproducibility, are given in Appendix B.

3.3 Trial design and statistical analysis

For a given random seed, the exact sequence of trials – subject, distractors, order – is identical across all three quantization levels, making every FP16-vs-INT4 (or FP16-vs-INT8) comparison a paired comparison at the trial level (verified with zero mismatches across 23,075 unique trials). The exact per-level trial budget and further detail on the paired design are given in Appendix C.

We report McNemar’s exact test as our primary significance test, exploiting this trial-level pairing, at each model’s most-powered interference level (selected for being farthest from ceiling/floor effects); we retain the unpaired Fisher’s exact test and a pooled Mantel–Haenszel test as more conservative cross-checks.

Table 1: Overall accuracy (all attributes and interference levels pooled) by model and quantization.

Model	FP16	INT8	INT4
Qwen2.5-7B-Instruct	97.6%	97.6%	96.5%
Mistral-7B-Instruct-v0.3	75.6%	74.4%	72.4%
Phi-3.5-mini-instruct	57.2%	53.0%	59.5%

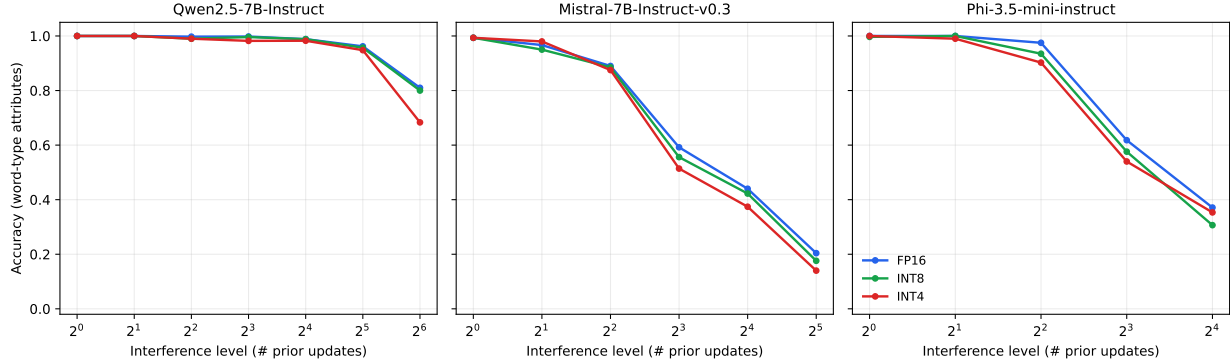


Figure 2: Word-type (semantic interference) accuracy vs. interference level, by model and quantization precision. Shaded region / error bars: omitted for clarity in the line plot; see Table 2 for exact accuracy and significance at each model’s key level.

To confirm the effect is not an artifact of level selection, we additionally fit a mixed-effects logistic regression using all eight interference levels simultaneously, with random intercepts for model, seed, and attribute, and a random-slopes robustness variant. We separately classify every incorrect word-type response as a same-key intrusion or non-matching error, and stratify the numeric-attribute comparison by token length to rule out a tokenization confound. Full formulas, model specifications, and robustness details are given in Appendix D.

4 Results

4.1 Overall accuracy

Table 1 shows overall accuracy (all attributes, all interference levels pooled) by model and quantization. Aggregate accuracy differences between FP16 and INT4 are small in absolute terms (1.1–2.4 percentage points) – consistent with the standard finding in the quantization literature that aggregate benchmarks are largely insensitive to 4-bit quantization. This is precisely why the effect we report below would not be visible from aggregate accuracy alone: it is concentrated in a specific, high-interference regime that aggregate metrics average away.

4.2 Word-type accuracy declines sharply under INT4 at high interference

Figure 2 shows word-type accuracy as a function of interference level. All three models show the expected log-linear-style decline in accuracy as interference increases [5]; critically, the INT4 curve separates visibly from FP16/INT8 at high interference in every model.

Table 2 reports the quantitative comparison at each model’s most-powered interference level (Section 3.3), using McNemar’s paired test (our primary test; Section 3.3) on the discordant trial pairs.

Table 2: FP16 vs. INT4 accuracy at each model’s most-powered high-interference level (word-type attributes), with McNemar’s paired test. n_{10} = trials correct under FP16 but not INT4; n_{01} = the reverse; $OR = n_{10}/n_{01}$. n = paired trials (pooled across 5 seeds). All three models show the same direction of effect in every individual seed (15/15 seed-comparisons). The unpaired Fisher’s exact test on the same cells agrees in direction and significance but is markedly less powered ($p = 0.0005, 0.0012, 0.015$ respectively), as expected once the trial-level pairing is discarded.

Model	Level	FP16	INT4	n_{10} / n_{01}	OR	p (McNemar)
Qwen2.5-7B-Instruct	64	81.0%	68.3%	40 / 2	20.00	4.1×10^{-10}
Mistral-7B-Instruct-v0.3	16	44.0%	37.4%	129 / 50	2.58	5.5×10^{-9}
Phi-3.5-mini-instruct	8	61.8%	54.0%	54 / 15	3.60	2.6×10^{-6}

Table 3: FP16 vs. INT8 accuracy at the same levels as Table 2 (word-type attributes), McNemar’s paired test. The unpaired Fisher’s exact test at these same cells gives $p = 0.837, 0.410, 0.197$ respectively (all n.s.) – the paired test recovers a real effect in two of the three models that the unpaired test misses.

Model	Level	FP16	INT8	n_{10} / n_{01}	OR	p (McNemar)
Qwen2.5-7B-Instruct	64	81.0%	80.0%	10 / 7	1.43	0.629
Mistral-7B-Instruct-v0.3	16	44.0%	42.2%	46 / 25	1.84	0.018
Phi-3.5-mini-instruct	8	61.8%	57.6%	35 / 14	2.50	0.004

Pooled unpaired cross-check. A Mantel–Haenszel test combining all three models’ unpaired 2×2 tables yields a common odds ratio of 1.40 (95% CI [1.23, 1.59]) and $p = 3.5 \times 10^{-7}$. This is consistent with, but considerably more conservative than, the paired McNemar result above, exactly as expected: pooling ignores the trial-level pairing entirely, so it represents a lower bound on how confident the paired design actually lets us be.

Revised INT8 picture: paired testing reveals a real, smaller effect in two of three models. Table 3 repeats the same paired analysis for FP16 vs. INT8. Under the unpaired Fisher’s test (as originally reported), all three models showed confidence intervals crossing 1 ($p = 0.837, 0.410, 0.197$), suggesting INT8 was statistically indistinguishable from FP16 – a clean “cliff” pattern ($FP16 \approx INT8 \gg INT4$). McNemar’s paired test tells a different story: Qwen remains a clean $FP16 \approx INT8$ case ($p = 0.629$), but Mistral and Phi both show a real, if smaller-magnitude, INT8 accuracy penalty ($p = 0.018$ and $p = 0.004$ respectively) that the unpaired test lacked the power to detect. We take this as the more trustworthy result, since it is the appropriately-matched test for our paired design. The practical picture is therefore not a clean cliff but a graded one: INT4 carries a large, robust penalty in every model (OR 2.6–20.0); INT8 carries a smaller but still real penalty in two of three models (OR 1.8–2.5), with Qwen as the one model where INT8 appears safe. Phi additionally shows one non-monotonic condition at $k = 16$, reported rather than omitted in Appendix E. INT8 should therefore not be assumed safe by default for every model.

4.3 A standardized summary metric: Interference Endurance Score (IES)

To enable standardized comparison with the PI-LLM literature and to summarize the quantization penalty in a single number, we compute the Interference Endurance Score (IES; 5): the area under the accuracy-vs-interference-level curve, integrated over a $\log_2$ -scaled interference axis. We report a normalized variant, dividing by the tested $\log_2$ range, so IES falls on the same $[0, 1]$ scale as accuracy and remains comparable across models even though Mistral and Phi’s smaller tokenizer-filtered word-type vocabularies cap their

Table 4: Interference Endurance Score (IES; area under the accuracy-vs- $\log_2$ -interference-level curve, normalized to $[0, 1]$) by model and quantization, word-type attributes. % lost is relative to FP16. Tested interference-level ranges differ by model due to the vocabulary-size cap in Table 7.

Model	Levels tested	FP16	INT8	INT4	INT4 % IES lost
Qwen2.5-7B-Instruct	1–64 (7 levels)	0.975	0.972	0.957	1.82%
Mistral-7B-Instruct-v0.3	1–32 (6 levels)	0.698	0.680	0.662	5.09%
Phi-3.5-mini-instruct	1–16 (5 levels)	0.820	0.791	0.777	5.16%

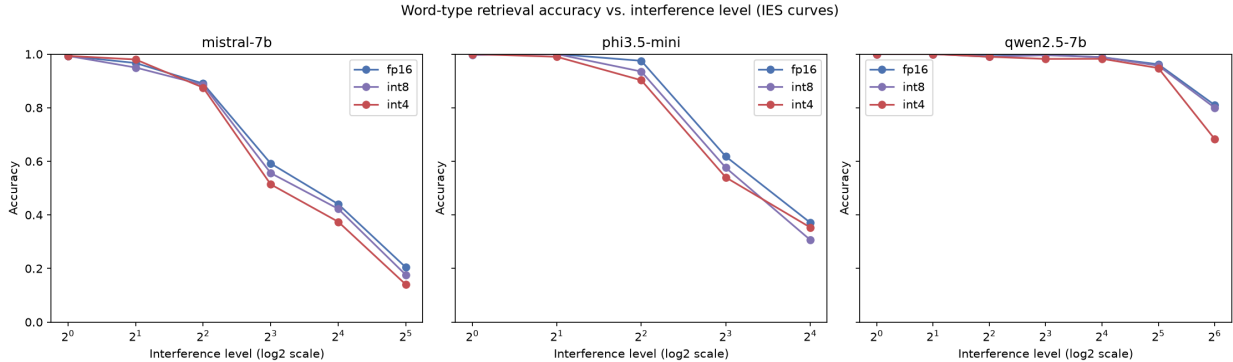


Figure 3: Accuracy vs. $\log_2$ -scaled interference level, FP16/INT8/INT4 overlaid per model – the curve underlying the IES computation in Table 4. Illustrates the cliff-dilution effect discussed in the text: Qwen’s near-flat curve through level 32 versus Mistral’s and Phi’s more gradual declines.

maximum tested interference level below Qwen’s (Table 7). IES is computed directly from the existing trial-level dataset; no new experiments were required.

Table 4 reports IES and the resulting quantization penalty for each model. INT4 quantization reduces IES relative to FP16 by 1.8% (Qwen2.5-7B), 5.1% (Mistral-7B), and 5.2% (Phi-3.5-mini) on word-type attributes, with INT8 falling consistently between FP16 and INT4 in every model – monotonic in the expected direction ($\text{FP16} \geq \text{INT8} \geq \text{INT4}$).

IES understates, rather than contradicts, the paper’s cliff finding for Qwen. Qwen’s accuracy remains near ceiling ($\geq 96\%$) through interference level 32 under every precision and only collapses sharply at level 64 (Table 2: 81.0% $\rightarrow$ 68.3%, a 13-point drop). Because IES averages over the full tested range and most of that range is saturated and identical across quantization levels for Qwen, this large, practically important local effect is diluted to a 1.8% aggregate IES loss. We therefore report IES alongside, not in place of, the per-level cliff analysis (Table 2): Mistral and Phi show more gradual decline starting at lower interference levels, so their IES penalties more directly reflect the accuracy curves in Figure 2, while Qwen’s small IES penalty specifically should not be read as evidence against its cliff.

Word-vs-numeric IES comparisons (Appendix E) partially corroborate the specificity result of Section 4.5: Mistral’s INT4 penalty is larger under word-type interference (5.1%) than numeric (2.6%), consistent with the paper’s central claim, though the pattern is not uniform across all three models.

4.4 Mixed-effects regression confirms the effect using all interference levels

The per-model analysis above necessarily selects one level per model for statistical power. To confirm this is not an artifact of that selection, Table 5 reports the mixed-effects logistic regression described in

Table 5: Mixed-effects logistic regression: $\text{correct} \sim \text{quant_mode} * \log_2(\text{level}) + (1 \mid \text{model}) + (1 \mid \text{seed}) + (1 \mid \text{attribute})$. FP16 is the reference level. The key row in each panel is the $\text{INT4} \times \log$ -interference-level interaction: negative for word-type, positive for numeric – opposite signs, both highly significant, even after accounting for attribute-level difficulty heterogeneity via the added random intercept.

Term	Coef. (β)	OR	95% CI	z (p)
<i>Word-type attributes (main hypothesis)</i>				
int8 (main effect)	−0.256	0.77	[0.73, 0.82]	−7.93 (2.2×10^{-15})
int4 (main effect)	−0.143	0.87	[0.81, 0.92]	−4.44 (9.1×10^{-6})
int8 $\times \log_2(\text{level})$	+0.023	1.02	[1.01, 1.04]	2.73 (0.0063)
int4 $\times \log_2(\text{level})$	−0.046	0.96	[0.94, 0.97]	−5.43 (5.7×10^{-8})
<i>Numeric attributes (specificity control)</i>				
int8 (main effect)	−0.213	0.81	[0.77, 0.85]	−8.52 ($< 10^{-16}$)
int4 (main effect)	−0.190	0.83	[0.79, 0.87]	−7.45 (9.2×10^{-14})
int8 $\times \log_2(\text{level})$	+0.013	1.01	[1.00, 1.02]	2.81 (0.0049)
int4 $\times \log_2(\text{level})$	+0.053	1.05	[1.04, 1.06]	11.05 ($< 10^{-16}$)

Section 3.3, fit separately on word-type and numeric trials, using *all* eight interference levels and all three models simultaneously.

For word-type attributes, the $\text{INT4} \times \log$ -interference-level interaction is negative and highly significant: the INT4 accuracy penalty relative to FP16 grows as interference increases. For numeric attributes, the same interaction term is *positive* and equally significant – INT4’s relative accuracy, if anything, improves slightly as numeric interference increases. This is a stronger specificity result than “no effect on the control condition”: the effect runs in the opposite direction, which is difficult to explain under any account where quantization simply injects generic noise into retrieval. This pattern is robust to adding random slopes for interference by model and attribute, and the main-effect coefficients (which describe the quantization gap only at the lowest interference level, $k = 1$, near ceiling for all conditions) should not be read as practically meaningful on their own; both points are detailed further in Appendix E.

4.5 Specificity: opposite-signed effect on numeric (non-semantic) interference

Figure 4 shows the numeric-attribute (control) accuracy-by-level curves. Baseline interference is present here too: pooled across all three models, accuracy declines from $\sim 100\%$ to $\sim 65\text{--}70\%$ as k grows to its highest tested values, confirming the task design elicits genuine interference-driven decline even for arbitrary values. This pooled figure, however, masks substantial per-model heterogeneity: Qwen remains comparatively robust even at high k (numeric accuracy stays above 90% through most of the tested range), while Mistral and Phi decline much more steeply. This is exactly why the mixed-effects model (Section 4.4) includes a random intercept for model: it separates each model’s overall difficulty level from the quantization-by-interference interaction we are actually testing.

At the three highest numeric interference levels (32, 64, 96, pooled across all three models), FP16 vs. INT4 differences are individually not significant ($p = 0.225$, $p = 0.123$, $p = 0.194$ respectively) and not consistently signed in FP16’s favor – consistent with, though less precise than, the mixed model’s positive interaction coefficient. Together, these results support the paper’s central claim: 4-bit quantization does not generically add retrieval noise that would degrade any interference-heavy task equally. It selectively erodes whatever mechanism resists *semantically confusable* distractors, while leaving resistance to arbitrary, non-confusable distractors intact – and if anything behaves oppositely there. Because the numeric vocabulary

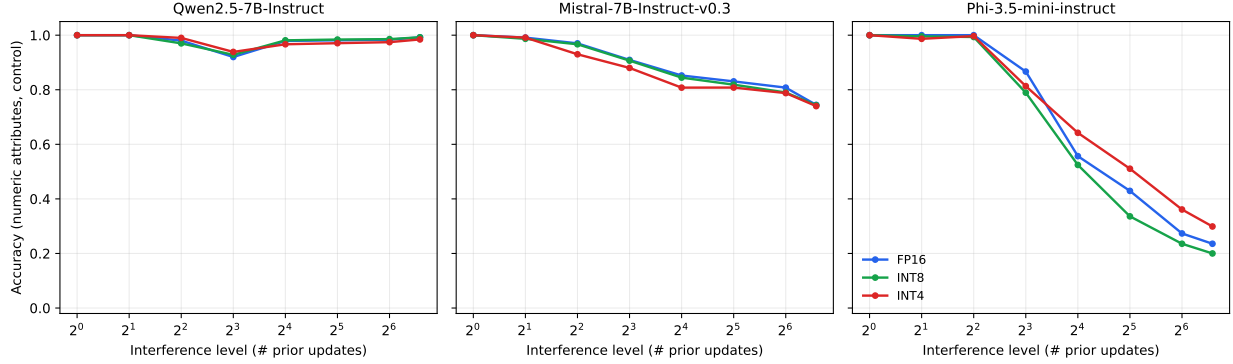


Figure 4: Numeric-attribute (control) accuracy vs. interference level, by model and quantization precision. Baseline interference is present, but no consistent quantization effect emerges at any level, in contrast to Figure 2.

is not tokenizer-filtered to be single-token, unlike the word-type pools, we separately verified that token-length composition is flat across interference levels and that the specificity result survives stratifying by token length (Appendix E), ruling out a tokenization confound as the explanation.

4.6 Mechanism: quantization increases the rate of same-key intrusions

The results above establish that INT4 quantization lowers retrieval accuracy under semantic interference, but accuracy alone does not show *how* the model fails. We reconstruct the exact ordered sequence of overwritten distractor values for every trial (deterministic given the seed; see Appendix D) and classify every incorrect word-type response as either a *same-key intrusion* – the extracted answer exactly matches one of the $k - 1$ earlier, overwritten values for that trial – or a non-matching error.

The overwhelming majority of errors are same-key intrusions at every precision level: 99.9% at FP16 (2,007/2,009 errors), 99.9% at INT8 (2,167/2,170), and 99.8% at INT4 (2,297/2,302). This confirms that when these models fail on this task, they are not answering at random – they are retrieving a real but stale value, consistent with the PI-LLM literature’s framing of this failure mode [5].

Critically, the *rate* of same-key intrusions, expressed as a fraction of *all* trials (not just errors), rises monotonically and significantly with more aggressive quantization: 21.5% of all word-type trials at FP16, 23.2% at INT8, and 24.6% at INT4 (Figure 6). A paired McNemar’s test on the FP16-vs-INT4 intrusion indicator, pooled across all models, seeds, attributes, and interference levels, confirms this is significant (OR = 0.84, 95% CI [0.78, 0.90], $p = 4.8 \times 10^{-7}$). Intrusion *recency* – how many positions back the intruding value came from – is comparable across precision levels (mean position 13.8 at FP16, 13.2 at INT8, 13.2 at INT4), suggesting quantization primarily changes *how often* the model confuses an earlier value for the current one, not *which* earlier value it confuses.

This is direct behavioral evidence for the mechanism this paper proposes in Section 5: quantization does not add generic answer noise, it specifically increases the rate at which semantically similar candidate values are confused with one another, which is exactly the signature that a coarsening of representational precision among semantically similar candidates – rather than a general accuracy loss – would predict.

4.7 Isolating the contribution of the output projection (lm_head)

A reviewer asked how much of the INT4 effect reported above is attributable to quantizing the output/vocabulary projection (lm_head) specifically, versus the transformer backbone. We address this with a follow-up

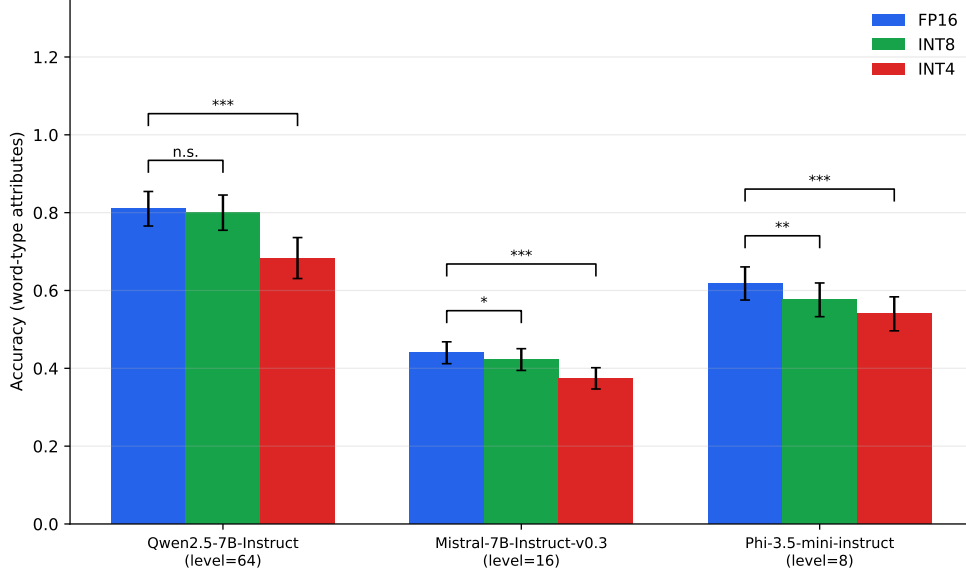


Figure 5: Accuracy at each model’s key high-interference level, word-type attributes only, with 95% confidence intervals. Brackets denote significance via McNemar’s paired test (our primary test, Section 3.3): the lower bracket compares FP16 vs. INT8, the upper bracket FP16 vs. INT4. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, n.s. not significant. Note the INT8 bracket is now significant for Mistral and Phi – the paired test recovers an effect the unpaired Fisher’s test (used in an earlier version of this analysis) missed.

ablation, restricted to word-type attributes at each model’s key high-interference level, using the same 5 seeds and identical environment as the main experiments. New trials for this ablation were generated to be paired by construction with the existing dataset, though this pairing was not re-verified with the same exhaustive, zero-mismatch check used for the main dataset; we treat this ablation’s paired comparisons accordingly as a well-motivated but slightly less rigorously verified companion to the paper’s primary paired results.

As noted in Section 3.2, our original INT4 condition did not, in practice, quantize `lm_head` at all, which required a methodological correction before the intended comparison could be run; the full story, including how this was diagnosed and corrected, is given in Appendix F. The corrected ablation condition verifiably quantizes `lm_head` on top of the existing backbone-only INT4 model and differs from it on 4.0% of paired trials (203/5,050), confirming the two are genuinely different models.

Results. Table 6 compares accuracy under backbone-only INT4 (the condition used throughout the paper) against INT4 with `lm_head` also quantized, at each model’s key interference level. Additionally quantizing `lm_head` produces no statistically detectable change in accuracy in any of the three models (paired McNemar’s test, all three comparisons $p = 1.0$, with only 3–4 discordant trials out of 300–1,200 paired trials in either direction). Decomposing each model’s total FP16→INT4(+`lm_head`) accuracy gap into a backbone-attributable share and an `lm_head`-attributable share shows that quantizing `lm_head` accounts for at most a few percentage points of the total effect: 2.5% for Qwen2.5-7B, 1.3% for Mistral-7B, and 0.0% for Phi-3.5-mini, none of which are statistically distinguishable from zero given the above. The underlying FP16→INT4 effect itself remains highly significant with `lm_head` quantized in every model (the `lm_head`-quantized condition vs. FP16: $p = 8.7 \times 10^{-10}$, 3.2×10^{-9} , and 3.8×10^{-6} for Qwen, Mistral, and Phi respectively).

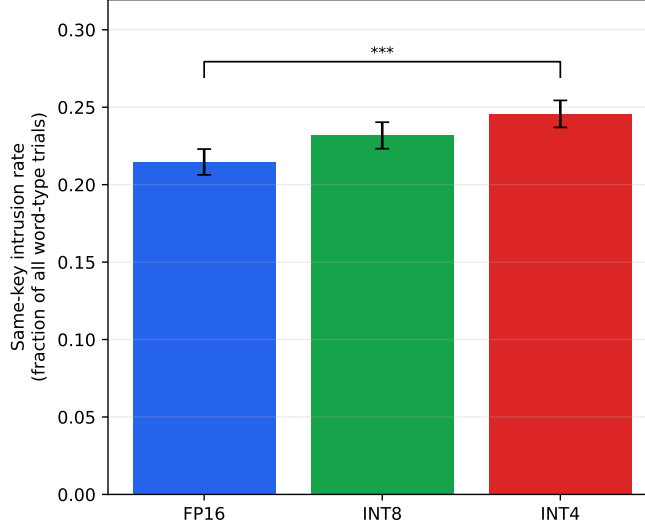


Figure 6: Same-key intrusion rate (fraction of all word-type trials whose wrong answer exactly matches an earlier overwritten value), by quantization precision, pooled across all three models, seeds, attributes, and interference levels. Error bars: 95% CI. Bracket: McNemar’s paired test, FP16 vs. INT4 ($p = 4.8 \times 10^{-7}$).

Table 6: Effect of additionally quantizing `lm_head` on top of the backbone-only INT4 condition, word-type attributes, at each model’s key interference level (same levels as Table 2). “INT4 (backbone)” is the condition used throughout the paper; “INT4 (+`lm_head`)” is the corrected ablation condition with `lm_head` also quantized. p -values are from McNemar’s paired test comparing the two conditions directly; none reach significance.

Model	Level	n	INT4 (backbone)	INT4 (+ <code>lm_head</code>)	<code>lm_head</code> share of gap	p
Qwen2.5-7B-Instruct	64	300	68.3%	68.0%	2.5%	1.0
Mistral-7B-Instruct-v0.3	16	1,200	37.4%	37.3%	1.3%	1.0
Phi-3.5-mini-instruct	8	500	54.0%	54.0%	0.0%	1.0

Interpretation. Across all three models, this ablation gives a consistent answer: the proactive-interference penalty under INT4 quantization is backbone-driven, not an artifact of quantizing the output projection. This is consistent with, and lends additional support to, the representational-coarsening mechanism proposed in Section 5 and directly evidenced by the same-key intrusion analysis (Section 4.6): both point toward the effect originating in the residual stream / backbone representations rather than in the final projection to vocabulary logits.

5 Discussion

Five findings together paint a consistent picture. First, the effect is real and generalizes: all three architecturally distinct models show the same direction of effect, every individual seed agrees with the aggregate direction (15/15), and our primary paired test (McNemar’s) is significant in every model ($p \leq 2.6 \times 10^{-6}$), with a more conservative unpaired pooled test agreeing ($p = 3.5 \times 10^{-7}$, common OR 1.40, 95% CI [1.23, 1.59]). Second, the effect is not an artifact of selecting a favorable interference level per model, nor of attribute-level difficulty differences, nor of unmodeled model- or attribute-specific interference trends:

a mixed-effects regression using all eight levels, all three models, and a random intercept per attribute confirms the INT4 penalty grows with interference ($\beta = -0.046$, $p = 5.7 \times 10^{-8}$), and this survives essentially unchanged when random slopes for interference by model and attribute are added ($\beta = -0.048$, $p = 1.9 \times 10^{-8}$). Third, the effect is specific: it appears under semantic (word-type) interference and runs in the *opposite* direction under numeric interference ($\beta = +0.053$, $p < 10^{-16}$), a result that survives stratifying by the numeric values’ token length (ruling out a tokenization confound as the explanation) – which together argue against a generic “quantization adds noise” explanation and toward a more specific mechanism – e.g. that 4-bit weight quantization coarsens the representational precision needed to keep semantically similar candidate values distinct in the residual stream, without similarly affecting representations of arbitrary, mutually dissimilar numeric tokens. Fourth, this proposed mechanism is not merely inferred from the accuracy pattern but directly observed in the errors themselves: virtually all incorrect responses (99.8–99.9%, at every precision level) are same-key intrusions rather than unrelated answers, and the *rate* of these intrusions rises significantly under INT4 relative to FP16 (24.6% vs. 21.5% of all trials, $p = 4.8 \times 10^{-7}$) – direct behavioral evidence that quantization specifically increases confusability among semantically similar candidates, rather than merely adding generic answer noise. A follow-up ablation isolating `lm_head` (Section 4.7) reinforces this same residual-stream account from a different angle: additionally quantizing the output projection on top of the backbone adds no statistically detectable effect in any of the three models (all $p = 1.0$), and accounts for at most 2.5% of the total FP16→INT4 gap where `lm_head` is quantized, so the effect is attributable to the quantized backbone rather than the final vocabulary projection. Fifth, the effect is graded rather than a clean cliff: our primary paired test shows INT4 carries a large, robust penalty in every model (OR 2.6–20.0), while INT8 carries a smaller but still real penalty in two of three models (Mistral, Phi; OR 1.8–2.5) that an unpaired test lacks the power to detect, with Qwen as the one model where INT8 remains statistically indistinguishable from FP16. INT8 should therefore not be assumed uniformly safe, though it remains meaningfully less risky than INT4 across the board. This narrower, task- and precision-specific pattern is also consistent with the broader theme identified by PTQ-Bench-style systematic benchmarking [9] and by cross-domain evidence from quantized world models [12]: quantization failure modes are frequently task-specific and component-specific rather than a simple monotonic function of bitwidth, so aggregate accuracy-vs-bitwidth summaries – even comprehensive ones – can miss behaviorally important failures that only targeted evaluation surfaces. A standardized summary statistic (IES; Section 4.3) corroborates this graded picture while also illustrating its own limits: it confirms a real, monotonic $\text{FP16} \geq \text{INT8} \geq \text{INT4}$ quantization penalty in every model, but substantially understates Qwen’s cliff specifically, since averaging over the full interference range dilutes a large effect concentrated at a single high interference level – underscoring why we report both the standardized metric and the per-level analysis rather than either alone.

Practical implications. Applications that rely on 4-bit quantized models to track frequently-updated, semantically dense state (e.g. an assistant tracking a user’s evolving preferences across a long conversation, or a system tracking the current status of many similar entities) may be more exposed to retrieval failures than aggregate benchmark scores would suggest. INT8 quantization remains a meaningfully safer choice than INT4 for such applications – its accuracy penalty is smaller in every model we tested, and undetectable in one of three – but our paired re-analysis indicates it is not risk-free either. Practitioners should not treat INT8 as categorically safe by default for interference-heavy retrieval tasks, particularly for models resembling Mistral or Phi in this study, even though both quantization levels are typically treated as “safe” compression levels in general-purpose benchmarking.

6 Limitations and Future Work

This work establishes a robust, multiply-verified effect within a specific and well-controlled scope, which also defines the natural directions for extending it. Our quantization coverage is bitsandbytes-specific (LLM.int8/NF4); as Section 4.7 shows, this concretely means the INT4 effect we report is a backbone effect, since the library leaves `lm_head` at full precision by default and a forced-quantization ablation found no additional penalty from including it. Extending this evaluation to calibration-based quantizers (GPTQ, AWQ) and newer formats (EXL2, HQQ) – several of which explicitly minimize output divergence from the unquantized model and may handle `lm_head` differently by default – would clarify whether the effect is specific to bitsandbytes’ NF4 scheme or reflects a more general cost of aggressive weight compression; a PTQ-Bench-style [9] sweep across quantizer families would be a natural vehicle for this and was scoped for the present revision but not yet completed. Activation and KV-cache quantization, increasingly common in production serving stacks, are likewise promising directions we have not yet tested.

Our numeric control condition uses a vocabulary that is not tokenizer-filtered to be single-token, unlike the word-type pools; we tested this directly (Section 4.5) and found token-length composition flat across interference levels with the specificity result holding in the dominant token-length stratum, which substantially de-risks this design choice without fully replacing a prospectively filtered design. Our prompts are also short-to-medium (at most a few hundred tokens), so whether the effect strengthens or attenuates in the 8k+-token long-context regime is an open, practically important question – one that prior work on quantized long-context retrieval [13] suggests could go either way and would be worth testing directly with our paradigm.

The mechanism we propose – that quantization coarsens representational precision among semantically similar candidates – is now supported by two independent lines of behavioral evidence (the same-key intrusion analysis of Section 4.6 and the backbone-localization ablation of Section 4.7), but neither constitutes a representational-level test. Probing model internals directly (attention patterns, residual-stream precision, activation distributions at the layers implicated in candidate-value tracking) is a natural next step to move this from a well-evidenced interpretation to a directly tested claim. Finally, extending the model pool beyond the 3.8–7B range tested here to larger and calibration-quantized-by-default models, and testing whether chain-of-thought prompting or alternative decoding strategies (only greedy decoding is tested here) interact with quantization, would clarify how broadly this effect generalizes.

7 Conclusion

We show that bitsandbytes 4-bit post-training quantization significantly degrades LLM retrieval accuracy under semantic proactive interference, across three architecturally distinct open models, confirmed by per-model McNemar’s paired tests at high-power interference levels (our primary test, exploiting the trial-level pairing built into the experimental design), by a more conservative unpaired pooled test, and by a mixed-effects regression using all interference levels simultaneously – with random intercepts for model, seed, and attribute, and, in a robustness check, random slopes for interference by model and attribute as well. This effect reverses sign under arbitrary (non-semantic) interference, a specificity result that survives stratifying by numeric token length, arguing against a generic quantization-noise account. A trial-level error analysis further shows this specificity at the mechanism level: quantization significantly raises the rate at which wrong answers are same-key intrusions on an earlier, overwritten value, rather than simply raising the overall error rate. A follow-up ablation further localizes the effect within the model: quantizing the output projection (`lm_head`) on top of the backbone adds no detectable effect beyond what backbone-only INT4 already produces, indicating the penalty originates in the quantized transformer backbone rather than the final vocabulary projection. The paired re-analysis also revises the paper’s original “cliff” narrative: INT4

carries a large, robust penalty in every model, while INT8 – often assumed safe – carries a smaller but real penalty in two of the three models once the appropriately-matched test is used. The overall effect is invisible to aggregate benchmark accuracy, which differs by only 1–2 percentage points between FP16 and INT4 in our data, yet reaches 10–13 percentage points at the specific high-interference, semantically-confusable conditions where it matters. All experiments were run on a single consumer-grade GPU, which we hope makes this kind of behavioral, non-aggregate evaluation more approachable for other researchers working under similar hardware constraints.

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A Full Task Design: Prompts, Scoring, and Vocabulary Construction

Prompt template. Each trial consists of a sequence of natural-language update statements followed by a retrieval question. Given an interference level k , exactly k update statements are generated for the same subject and the same attribute, with each statement overwriting the previous value. The updates appear in chronological order, and the final update always contains the ground-truth value. The retrieval question is appended after a blank line and explicitly instructs the model to report only the most recent value. For example, a word-type trial takes the following form:

Example prompt (word-type trial with k equal to 3)

Adam’s mood is now happy.
Adam’s mood is now nervous.
Adam’s mood is now calm.

Based only on the most recent update above, what is Adam’s mood right now? Reply with a single word and nothing else.

Numeric attributes use the same structure but replace the update template and question accordingly (e.g., temperature or stock price) while requesting only the final numeric value. Thus every prompt follows the general pattern

$$\underbrace{\text{Update}_1, \text{Update}_2, \dots, \text{Update}_k}_{\text{chronological overwrites}} \rightarrow \text{Retrieval Question}$$

where the only correct answer is the value introduced in the final update, and the prompt structure remains identical across all trials. Prompts are provided to each model through its native instruction-chat template (e.g., Qwen, Mistral, and Phi chat formats) using the tokenizer’s built-in function, as a single user message with no system prompt.

Scoring and exact-match criteria. Models are prompted to answer with a single word (word-type attributes) or a single number (numeric attributes) and nothing else, and generation is capped at 12 new tokens. The raw response is lowercased and stripped of leading/trailing whitespace and a single trailing period, if present. For word-type attributes, we take the first contiguous alphabetic substring in the cleaned response (via a simple regular expression) and compare it against the gold value, also lowercased; a trial is scored correct only on an exact string match, with no partial-credit or fuzzy matching. For numeric attributes, we take the first contiguous run of digits (optionally signed) in the cleaned response and compare it against the gold integer as a string. Because generation is greedy and prompts explicitly instruct a single-token-style answer, the overwhelming majority of responses consist of exactly the target word or number with no additional content; the regular-expression extraction step exists primarily to strip a stray trailing period or an occasional short preamble rather than to resolve ambiguous outputs. We apply no normalization beyond casing, whitespace, and the trailing-period strip described above (e.g. no stemming, no synonym matching, no numeric tolerance window), so scoring is deliberately conservative in the same way across all three quantization levels.

Tokenizer-verified vocabulary construction. For word-type attributes, each interference level k requires k unique values sampled without replacement from that attribute’s vocabulary pool. A naive approach risks two confounds as pools are made large enough to support high k : (1) word rarity drift (later-added words being less common), and (2) token-length drift (later-added words spanning more subword tokens). Both would let interference level and item difficulty covary, contaminating any accuracy-vs-level relationship. We mitigate this by fixing one large candidate list per category (80–140 common, everyday words) and filtering it, per tokenizer, to keep only words that occupy exactly one token *as they actually appear in the trial template*. We determine this by tokenizing the fully instantiated template string and checking, via the tokenizer’s character-offset mapping, how many resulting tokens overlap the candidate word’s character span. This check is robust across tokenizer families: byte-level BPE tokenizers (e.g. Qwen) fuse a leading space into the following word’s token, while SentencePiece-based tokenizers (e.g. Mistral, Phi) may add a constant leading token independent of content; both behaviors are correctly handled by reading offsets directly off the real tokenized sentence rather than inferring token count arithmetically from separately-encoded fragments.¹ This yields a fixed, per-model, per-category vocabulary pool whose maximum usable interference level differs by model (Table 7) but whose quality does not vary with k : every level for a given model/category samples from the same fixed, filtered pool.

Table 7: Maximum interference level tested per word-type attribute, by model (limited by tokenizer-filtered single-token vocabulary size).

Model	mood	favorite color	favorite animal	occupation
Qwen2.5-7B-Instruct	64	32	32	32
Mistral-7B-Instruct-v0.3	32	16	16	16
Phi-3.5-mini-instruct	16	16	8	16

Numeric attributes draw from a pool of ~ 990 unique integers per category and therefore support all tested interference levels for every model. Unlike the word-type pools, we do not explicitly filter numeric values to be single-token: integers in our range (3–998) can span one or two tokens depending on the tokenizer and value. This does not reintroduce the level-drift confound the word-type filtering was designed

¹An earlier version of our pipeline used exactly such an arithmetic approach (subtracting a “blank” baseline encoding) and produced silently corrupted vocabulary pools for one tokenizer family while appearing to work for another – we flag this as a methodological pitfall worth documenting for others building similar evaluations.

to prevent, because the numeric pool is fixed and level-invariant – every interference level samples from the same range of integers, so token-count composition does not systematically covary with k the way an insufficiently large word pool would. We nonetheless flag this as a minor asymmetry in rigor between the two conditions (Section 6).

B Reproducibility: Model Loading Configuration

All experiments were conducted on a single NVIDIA RTX 3090 GPU with 24 GB of VRAM. Unless otherwise specified, all models were loaded using the configuration summarized in Table 8. No additional dtype autocasting was applied beyond the compute dtype specified for INT4 quantization.

Table 8: Model loading and quantization configuration used in all experiments.

Setting	Value
GPU	NVIDIA RTX 3090 (24 GB VRAM)
Device map	<code>{"":0}</code>
Low CPU memory usage	True
INT8 configuration	
BitsAndBytes	<code>load_in_8bit=True</code>
LLM.int8 outlier threshold	6.0 (default)
<code>llm_int8_skip_modules</code>	None
INT4 configuration	
BitsAndBytes	<code>load_in_4bit=True</code>
Quantization type	NF4
Double quantization	Enabled
Compute dtype	FP16(<code>torch.float16</code>)
NF4 block size	64 (default)
<code>modules_to_not_convert</code>	None
Additional dtype autocasting	None

C Trial Budget, Seeds, and Paired Design

Each experimental condition (model $\times$ quantization $\times$ interference level $\times$ attribute) was evaluated using five independent random seeds, resulting in a total of 69,225 trials (28,050 word-type and 41,175 numeric) across 45 (model $\times$ quantization $\times$ seed) runs.

For a fixed (model, seed) pair, the complete sequence of generated trials (including subject names, distractor values, trial ordering, attributes, and interference levels) was identical across FP16, INT8, and INT4. Thus, each quantization level was evaluated on exactly the same set of prompts, with the model weights being the only difference between runs. Consequently, all FP16–INT8 and FP16–INT4 comparisons reported in this paper are performed using paired trials rather than independently sampled trials, eliminating variation in item difficulty as a potential confounding factor.

Across different seeds, subject names and distractor values were sampled independently from the same fixed, tokenizer-filtered vocabulary pool. Finally, the trial budget was intentionally weighted toward higher interference levels, where floor and ceiling effects are less likely to obscure the impact of quantization (Table 9).

Table 9: Trials per interference level, per (model, quantization, seed, attribute) cell.

Level k	1	2	4	8	16	32	64	96
Trials	15	15	20	25	60	50	60	60

D Statistical Methodology: Full Details

For each model, we identify the interference level with the greatest statistical power for detecting a quantization effect – i.e. farthest from ceiling (ceiling effects dominate at low k for all models) and farthest from floor (floor effects dominate at very high k for the weaker models): Qwen at $k = 64$, Mistral at $k = 16$, and Phi at $k = 8$. At each selected level we compare FP16 vs. INT4 accuracy via Fisher’s exact test and report the odds ratio with a 95% confidence interval (computed with statsmodels; 17), and separately confirm the direction of the effect (FP16 > INT4) is consistent across all 5 seeds individually. To test whether the effect generalizes across models rather than relying on three separate per-model tests, we compute a Mantel–Haenszel test [18], pooling the three models’ 2×2 tables into a single common odds ratio, confidence interval, and significance test that controls for model identity as a stratifying variable.

Second, and more importantly, because the per-model level selection above is chosen post-hoc for statistical power (a reasonable but auditable choice that a skeptical reader might still flag as researcher degrees of freedom), we additionally fit a mixed-effects logistic regression using *all* interference levels simultaneously, for every model and seed at once:

$$\text{correct} \sim \text{quant_mode} * \log_2(\text{interference_level}) + (1 \mid \text{model}) + (1 \mid \text{seed}) + (1 \mid \text{attribute})$$

fit via variational Bayes using a Bayesian mixed generalized linear model (statsmodels; 17), with FP16 as the reference level for quantization mode and random intercepts for model, seed, and attribute. The attribute-level random intercept is included because our word-type and numeric attributes differ substantially in baseline difficulty (e.g. mood is markedly harder than favorite animal at the same interference level); without it, this attribute-level heterogeneity could in principle be partly absorbed into the fixed effects and inflate their apparent significance. The key term is the quantization-mode $\times$ log-interference-level interaction: a significant negative coefficient for INT4 means the INT4 accuracy penalty relative to FP16 grows as interference increases, using the complete dataset rather than any selected subset. We fit this model separately for word-type and numeric attributes, since the specificity claim predicts opposite signs for the two.

Paired significance testing (McNemar’s test). The per-model comparisons above (Fisher’s exact test, Mantel–Haenszel) are unpaired: they compare aggregate correct/incorrect counts between two quantization levels without exploiting the fact that, by construction, the same trial content is presented to both. We verified this pairing holds exactly across our full dataset: for every one of the 23,075 unique (model, seed, attribute, level, trial) combinations, the logged subject name and gold value are identical across all three quantization levels, with zero mismatches. Given this, we report McNemar’s exact test as our primary per-model significance test at each model’s selected level: this test uses only the discordant pairs (trials on which the two compared quantization levels disagree) and is therefore substantially better powered than the marginal Fisher’s exact test, which discards the pairing entirely. We retain the unpaired Fisher’s-exact/Mantel–Haenszel results as a more conservative cross-check that does not depend on the pairing assumption.

Random-slopes robustness check. To rule out the possibility that a model-specific or attribute-specific interference-level slope is being absorbed into the fixed-effect interaction terms of the mixed model above, we refit it with two additional variance components: random slopes for $\log_2(\text{level})$ by model and by attribute, alongside the random intercepts already described.

Error-type classification (same-key intrusion analysis). Because trial generation is fully deterministic given its random seed, we reconstruct the exact ordered sequence of overwritten distractor values for every trial by replaying the seeded sampling procedure, and confirm this reconstruction exactly reproduces the logged subject name and gold value for all 23,075 unique trials, with zero mismatches. This lets us classify every incorrect response as a *same-key intrusion* (the extracted answer exactly matches one of the $k - 1$ overwritten values for that trial) or a non-matching error, and to compute an intrusion rate as a fraction of all trials, not just of errors.

Numeric token-length stratification. To address the token-length asymmetry between our tokenizer-filtered word-type pools and the unfiltered numeric pool, we separately tokenize every numeric value actually used in a trial, in its real template context, per model tokenizer, using the same offset-mapping method as the word-type filtering. We use this to check whether token-length composition drifts with interference level (which would reintroduce the confound the word-type filtering was designed to avoid) and to stratify the FP16-vs-INT4 numeric comparison by token length.

Interference Endurance Score (IES). Following Wang and Sun [5], we define IES as the area under the accuracy-vs-interference-level curve, computed via trapezoidal integration over a $\log_2$ -scaled interference axis. We report a normalized variant – raw AUC divided by the tested $\log_2$ range – so that IES falls on the same $[0, 1]$ scale as accuracy and remains comparable across models despite differing maximum tested interference levels (Table 7). IES is computed directly from the paper’s existing trial-level dataset (Section 3.3), requiring no new experiments.

E Additional Results and Robustness Checks

Phi non-monotonicity at $k = 16$. Table 10 shows the one condition in our data where INT8 accuracy falls below INT4. We report this rather than omit it; it does not change the paper’s overall conclusions, since Phi’s key-level comparison (Table 2, Table 3) is conducted at $k = 8$, not $k = 16$.

Table 10: Phi-3.5-mini-instruct at $k = 16$: the one condition in our data where INT8 accuracy falls below INT4, $n = 300$ per cell.

	FP16	INT8	INT4
Accuracy	37.1%	30.7%	35.3%

Interpreting the mixed-model main effects. Beyond the interaction terms reported in Table 5, the model also shows sizeable negative main effects for both INT8 and INT4 relative to FP16, in both word-type and numeric attributes. Because interference level is log-transformed and centered at $\log_2(1) = 0$, these main-effect coefficients describe the quantization gap specifically at the lowest interference level ($k = 1$), where accuracy is near ceiling for every model and condition (Table 1, Figure 2). A significant main effect alongside a near-ceiling raw accuracy simply reflects that logistic regression coefficients operate on the log-odds scale, where small absolute probability differences near 100% correspond to comparatively large log-odds differences; it should not be read as evidence of a practically meaningful gap at $k = 1$, where raw accuracy differences between quantization levels are well under one percentage point in our data. The interaction terms, not the main effects, are the theoretically relevant quantities for this paper’s hypothesis, since they capture how the quantization gap *changes* with interference rather than its value at one arbitrary reference level.

Robustness to random slopes. Table 5 includes random intercepts for model, seed, and attribute, but not random *slopes* for interference by model or attribute – leaving open the possibility that a model-specific or attribute-specific interference trend is being absorbed into the fixed-effect interaction terms. Refitting with random slopes for $\log_2(\text{level})$ by both model and attribute leaves the key interaction essentially unchanged in both panels: for word-type, the $\text{INT4} \times \log\text{-interference-level}$ interaction becomes $\beta = -0.048$, $\text{OR} = 0.95$, 95% CI [0.94, 0.97], $z = -5.62$, $p = 1.9 \times 10^{-8}$ (vs. $\beta = -0.046$, $p = 5.7 \times 10^{-8}$ without random slopes); for numeric, it becomes $\beta = +0.059$, $\text{OR} = 1.06$, 95% CI [1.05, 1.07], $z = 11.84$, $p < 10^{-16}$ (vs. $\beta = +0.053$, $p < 10^{-16}$). The opposite-signed specificity result – this paper’s central claim – is therefore not an artifact of unmodeled model- or attribute-specific interference trends. One secondary term does change: the numeric $\text{INT8} \times \log\text{-interference-level}$ interaction, weakly significant without random slopes ($p = 0.0049$), becomes non-significant with them ($\beta = +0.004$, $p = 0.369$); this term is not part of the main hypothesis (which concerns INT4), so it does not affect our conclusions.

Ruling out a token-length confound. Because the numeric pool is not tokenizer-filtered to be single-token, unlike the word-type pools, a skeptical reading of the specificity result is that it reflects token-length differences rather than semantic (non-)confusability. Two checks address this. First, the token-length composition of the numeric values actually used is essentially flat across all eight interference levels (dominated by 3-token numbers at 89.9–92.4% of trials at every level, with 2-token and 1-token values each maintaining a roughly constant share), confirming the numeric pool does not exhibit the level-drift confound the word-type filtering was designed to prevent. Second, stratifying the FP16-vs-INT4 comparison at high interference levels (32, 64, 96 pooled) by token length shows the specificity direction holds in every stratum and is individually significant in the dominant 3-token stratum, which accounts for >89% of all numeric trials: FP16 69.4% vs. INT4 71.2% ($n = 6,882$ pairs, $\text{OR} = 0.92$, 95% CI [0.85, 0.99], $p = 0.021$). The 1-token ($n = 32$) and 2-token ($n = 736$) strata point in the same direction but are individually underpowered ($\text{OR} = 0.86$, $p = 0.78$; $\text{OR} = 0.94$, $p = 0.57$). The numeric-attribute asymmetry flagged in Section 6 does not appear to drive the specificity result.

IES on numeric (control) attributes. Table 11 reports the same INT8/INT4 IES penalty computed on numeric attributes, for comparison with the word-type results in Section 4.3. The pattern only partially corroborates the specificity result of Section 4.5. For Mistral, the INT4 penalty is clearly larger under word-type interference (5.1%) than numeric (2.6%), consistent with the paper’s central claim. For Qwen, both penalties are small and comparable (1.8% vs. 0.05%), which is consistent with the cliff-dilution effect discussed above rather than with a genuine word/numeric asymmetry at this model’s tested levels. For Phi, INT4’s numeric IES is *higher* than FP16’s (−3.98%, i.e. INT4 nominally outperforms FP16 on the numeric control by this metric); this is a small-magnitude reversal consistent with the sign flip already established for numeric attributes in the mixed-effects analysis (Section 4.4), and we treat it as a null/mixed result for this model rather than evidence against the paper’s specificity claim. Phi’s INT8 numeric penalty (5.49%) is notably larger than its word-type INT8 penalty (3.53%); we have not independently traced this to a specific interference level and report it as an observed, unexplained asymmetry rather than folding it into the main narrative.

A methodological detail not independently re-verified. IES for word-type attributes pools across four attributes (mood, favorite color, favorite animal, occupation) whose tokenizer-filtered vocabulary caps differ within a model (Table 7: e.g. for Qwen, mood reaches $k = 64$ but occupation caps at $k = 32$). We did not verify how the underlying computation handles this asymmetry – whether the highest tested levels are pooled only across the attributes that support them, or some other convention – and report the IES values in Section 4.3 as provided, flagged here as an unverified detail rather than a confirmed part of the methodology.

Table 11: INT8/INT4 IES penalty (% lost relative to FP16) on numeric (control) attributes, for comparison with the word-type results in Table 4.

Model	INT8 % IES lost	INT4 % IES lost
Qwen2.5-7B-Instruct	−0.07%	0.05%
Mistral-7B-Instruct-v0.3	0.76%	2.60%
Phi-3.5-mini-instruct	5.49%	−3.98%

F lm_head Ablation: Additional Methodological Detail

Our first attempt at this ablation reran the INT4 configuration with the output projection (lm_head) explicitly excluded from quantization via a skip-list argument. This produced outputs identical to the original INT4 condition on all 5,050 paired trials, token for token. Inspecting the loaded model revealed why: under the installed library version, lm_head is left at full precision by default irrespective of this argument. This means the INT4 condition used throughout the paper never quantized lm_head in the first place – a property of the library’s defaults, not an error in our setup – so every INT4 result in the main text reflects backbone-only quantization.

To obtain the intended comparison, we instead manually converted lm_head to a genuine 4-bit layer on top of the standard INT4 model, matching the backbone’s quantization settings, and confirmed the conversion took effect before running any trials. This corrected condition differs from the original backbone-only INT4 condition on 4.0% of paired trials (203/5,050) – a small but genuine footprint, as expected from quantizing one additional layer, and the basis for the comparison reported in Section 4.7. Full implementation details are available in the released code repository.